Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune

Implementation Solution Guide

Version 1.0.0 · 2026-06-27

Contents

Executive Summary — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune 5

The problem . . . 5

The approach . . . 5

What you get . . . 5

The outcome . . . 6

1. Solution Overview — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune 6

What this solution delivers . . . 6

Devices in scope . . . 7

What is in scope vs. referenced only . . . 7

Critical accuracy note — get the Epic print specifics from your Epic team . . . 7

Why this approach . . . 7

How to use this package . . . 8

2. Requirements Guide — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune 8

R1. Microsoft Intune licensing and configuration + app rights . . . 8

R2. The standard clinical-workstation device group . . . 9

R3. Printer delivery method decided — Universal Print vs scripted install . . . 9

R4. Universal Print licensing and registration (only if using Universal Print) . . . 9

R5. Print queues / printer endpoints and drivers (only if using scripted install) . . . 9

R6. Clinical peripheral drivers/apps — scanners, signature pads, badge/proximity readers . . . 10

R7. Epic print mapping values — from Epic Technical Services / UserWeb . . . 10

R8. Windows security baseline (apply alongside this baseline) . . . 10

R9. Target operating system — Windows 10 / Windows 11 . . . 10

R10. A test clinical device (not in patient care), in a representative unit . . . 11

R11. Approved clinical change window and rollback plan . . . 11

R12. Imprivata present (only if badge readers feed SSO / badge-tap) . . . 11

Requirements checklist (use before standardizing) . . . 11

3. Implementation Guide — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune 12

Phase A — Assess . . . 12

Phase B — Standardize (the device baseline) . . . 13

Phase C — Deploy Peripherals / Printing . . . 14

Phase D — Pilot . . . 15

Phase E — Validate . . . 16

Phase F — Expand . . . 16
5. Testing & Validation — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune 17
Test 1 — Standardization baseline applied . . . 17
Test 2 — Power & display behavior correct . . . 17
Test 3 — Unit printers present in the correct order . . . 18
Test 4 — Default printer equals the unit default . . . 18
Test 5 — Clinical peripherals recognized . . . 18
Test 6 — Badge reader feeds SSO (only if using Imprivata, R12) . . . 18
Test 7 — Epic wristband prints to the wristband printer . . . 18
Test 8 — Epic script/prescription prints to the script printer . . . 18
Test 9 — Epic document prints to the document printer / default . . . 19
Test 10 — Repeatability across form factors . . . 19
Acceptance gate (must hold before Phase F expansion) . . . 19
6. Troubleshooting — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune 19
Standardization baseline . . . 20
Power / display (device sleeps in a patient room) . . . 20
Printers — presence and order . . . 20
Default printer wrong . . . 20
Universal Print vs scripted-install issues . . . 20
Clinical peripherals . . . 21
Epic mapping mismatches . . . 21
Drift and expansion . . . 21
7. Best Practices — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune 22
Naming and self-documentation . . . 22
Standardize through groups, not per-device clicks . . . 22
Printer delivery — choose deliberately, document the choice . . . 22
The default printer is a patient-safety control . . . 22
Peripherals: package once, deploy everywhere . . . 23
Respect the Epic / print-team boundary . . . 23
Clinical-environment safety . . . 23
Hold the line against drift . . . 23
8. Deliverables Summary — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune 24
Documentation . . . 24
PowerShell toolkit (`scripts\`) . . . 24

Visual asset 25
Named objects you create (reused verbatim across the docs and scripts) 25
Appendix A — PowerShell Scripts 26

Executive Summary — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune

The problem

Clinical Windows workstations that run Epic — nurse stations, exam-room PCs, medication-room and registration desks — drift apart over time. Each was imaged at a different point, joined by a different technician, and hand-configured with whatever printers and peripherals were near it that day. The result is a fleet where the same Epic workflow behaves differently from one room to the next: a wristband prints to the wrong tray, a script routes to a document printer three doors down, a signature pad is unrecognised at registration, and the workstation a float nurse sits at logs them out before they finish charting. Every inconsistency becomes a help-desk ticket, and in a clinical setting every ticket is a delay at the point of care.

This solution removes that drift. It defines a single, repeatable device baseline for clinical workstations and delivers it entirely through Microsoft Intune — configuration profiles, the settings catalog, power and display behaviour, and a documented reference to the organisation's Windows security baseline. On top of that baseline it builds the **clinical peripheral and printing layer** that Epic workflows depend on: context-aware printer deployment for wristband and label printers, prescription/script printers, and document printers; a correct **default printer per unit or department**; and the supporting peripherals — scanners, signature pads, and badge/proximity readers — installed and validated the same way on every device.

The approach

The work follows a disciplined, six-phase model — **Assess** → **Standardize** → **Deploy Peripherals/Printing** → **Pilot** → **Validate** → **Expand** — so that nothing reaches a live clinical floor before it has been proven on a pilot group. Devices are governed through a single Entra ID group, `EPIC – Clinical Workstations`, and every configuration profile, printer-deployment policy, and printer queue is named to a fixed convention so the fleet is self-documenting. Printer delivery is deliberate about the trade-off between **Universal Print** (cloud-native, no print server, requires the documented licensing and a registered printer or connector) and **scripted printer installation** (works with existing print servers and direct-IP queues), and the solution explains exactly when each applies.

Epic-internal values — the Epic Print Service / EPS configuration, queue names, and workflow-specific print settings — are owned by the customer's Epic Technical Services team and sourced from Epic UserWeb. This solution does not invent those values; it provides the standardization, deployment, and validation mechanics that carry them onto every workstation correctly and prove they landed.

What you get

A complete, white-label implementation package: a solution overview, a no-assumptions requirements specification (with a category-wide global-requirements companion), a step-by-step implementation guide with full Intune navigation paths, a testing and validation plan, a best-practices reference, a troubleshooting guide, and a deliverables summary. Five production-ready PowerShell scripts handle

discovery, per-device validation, fleet reporting, safe remediation, and printer-deployment mapping — all read-only or `-WhatIf` by default, so nothing changes in a clinical environment without an explicit, deliberate action.

The outcome

Every clinical workstation presents the same Epic environment: the same baseline, the same printers in the same order, the right default printer for the room, and working peripherals — verified, repeatable, and expandable unit by unit across the organisation.

1. Solution Overview — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune

This solution gives you a single, repeatable way to **standardize clinical Windows workstations for Epic workflows** using **Microsoft Intune** — a consistent device baseline (configuration profiles, the settings catalog, power and display behavior) plus the **clinical peripheral and printing layer** those workflows depend on: context-aware **printer deployment** (wristband/label, prescription/script, and document printers), a correct **default printer per unit**, and the supporting peripherals — scanners, signature pads, and badge/proximity readers — installed and validated the same way on every device.

The work follows the **standard deployment model**: **Assess** → **Standardize** → **Deploy Peripherals/Printing** → **Pilot** → **Validate** → **Expand**. Nothing reaches a live clinical floor before it has been proven on a pilot group.

What this solution delivers

- A **standardized device baseline** for clinical workstations, delivered through Intune

configuration profiles and the **settings catalog**: power and display behavior tuned for always-on shared clinical devices, lock/timeout behavior, and a documented reference to your organization's Windows security baseline — all bound to one named device group.

- A **context-aware printer-deployment layer** that puts the **right printers, in the right order,

with the right default**, on each workstation by unit/department — wristband/label printers, prescription/script printers, and document printers — delivered via** Universal Print **or** scripted printer installation**, whichever fits your print infrastructure.

- A **clinical peripheral layer**: document **scanners**, **signature pads** (registration /

consent), and **badge / proximity readers**, with their drivers/apps packaged and deployed through Intune and validated on every device.

- A **validation gate** (documented tests plus PowerShell scripts) that confirms the baseline

applied, the correct printers and default printer are present per unit, and the peripherals are recognized — before the rollout is allowed to expand.

- Five PowerShell tools (discovery, per-device validation, fleet reporting, safe remediation, and

printer-deployment mapping) to operate the standard day to day — all read-only or `-WhatIf` by default.

Devices in scope

- **Clinical workstations** — nurse stations, medication-room and registration desks.
- **Exam-room PCs** and other shared clinical endpoints.
- **Workstations-on-wheels (WOWs)** / mobile carts where they share the same Epic workflows.

These are typically **shared, multi-user, always-on** devices on a clinical network. That is exactly why a single named baseline, deterministic printer order, a correct per-unit default printer, and a validated pilot-then-expand rollout matter more here than on a standard office fleet.

What is in scope vs. referenced only

In scope (this solution): the Intune **device standardization** (configuration profiles, settings catalog, power/display), the **printer-deployment and default-printer-per-unit** layer (Universal Print vs scripted install), the **peripheral** (scanner / signature pad / badge reader) deployment, and the validation that proves all of it landed.

Referenced only (covered in depth by separate solutions):

- **Epic client packaging and lifecycle** (Hyperdrive/Hyperspace install, version pinning,

supersedence) — this solution standardizes the *workstation and its peripherals around* the Epic client, not the client package itself.

- **Imprivata** single sign-on / badge-tap / EPCS — referenced where a **badge/proximity reader**

must be recognized by the OS; the SSO configuration itself is a separate solution.

- **The Windows security baseline** — referenced as a named prerequisite to apply alongside this

baseline, not redefined here.

Critical accuracy note — get the Epic print specifics from your Epic team

The **Epic Print Service (EPS)** configuration, the **Epic print queue names**, the **workflow-specific print settings** (which queue a wristband, a script, or a document routes to), and the mapping of Epic departments to physical printers are defined in **Epic's customer-only documentation (UserWeb)** and by **your organization's Epic Technical Services / client systems team**. This solution **does not invent** those Epic-internal values — and you should not either.

Throughout the Implementation Guide you will see placeholders such as `<EPS_SERVER>`, `<EPIC_QUEUE_WRISTBAND>`, `<EPIC_QUEUE_SCRIPT>`, `<EPIC_QUEUE_DOC>`, and `<UNIT_DEFAULT_PRINTER>`. **Obtain the real values from your Epic team / Epic UserWeb** and substitute them consistently. The Intune mechanics around those values — configuration profiles, printer-deployment policy, Universal Print vs scripted install, driver packaging, and validation — are documented here exactly and accurately.

Why this approach

- **Clinical consistency at the point of care:** when every workstation in a unit presents the same

baseline, the same printers in the same order, and the right default printer for the room, an Epic workflow behaves identically wherever a clinician sits — fewer mis-prints, fewer logon surprises, fewer tickets.

- **Deterministic printing:** a wristband must print to the wristband printer and a script to the script printer, every time. Naming the printer-deployment policy and default per unit, and validating it, makes that deterministic rather than incidental.
- **Safe rollout:** pilot-then-expand with a named device group contains the blast radius of any baseline or printer change to a small, validated set of clinical endpoints before it reaches the floor.

How to use this package

Read the documents in order:

1. **02 – Requirements** and **global-requirements-healthcare-solutions** — confirm everything is in place before you configure anything.
1. **03 – Implementation** — standardize the baseline, deploy printers/peripherals, pilot, validate, then expand.
1. **05 – Testing** — the acceptance gate that must pass before expansion.
2. **07 – Best Practices**, **08 – Deliverables Summary**, and **06 – Troubleshooting** as needed.

2. Requirements Guide — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune

Verify **every** requirement below before you standardize, deploy printers, or deploy peripherals. Each entry follows the same pattern: **Description · Why required · How to verify · How to implement if missing.** Do **not** assume any prerequisite already exists.

> *Category-wide prerequisites (a tenant, Intune subscription, device enrollment, HIPAA / HITRUST change control, clinical maintenance windows, a test device, rollback posture, etc.) are documented once in* `global-requirements-healthcare-solutions.md`*. This guide lists what is* ***specific to*** *standardizing clinical workstations and their peripherals/printing.*

R1. Microsoft Intune licensing and configuration + app rights

Description: An Intune subscription and an operator with rights to create and assign **device configuration profiles** and **Windows apps** (Intune Administrator or, scoped, Policy/Profile Manager + Application Manager). **Why required:** The baseline is delivered as Intune **configuration profiles / settings catalog**, and peripherals/printer-install logic are delivered as **Win32 apps** or **platform scripts**; without these rights you cannot author or assign them. **How to verify:** `https://intune.microsoft.com` → **Devices** → **Configuration** shows **+ Create**, and **Apps** → **Windows** shows **+ Add**; Entra → **Roles and administrators** shows the operator's role. **How to implement if missing:** Assign an Intune-bearing license and the **Intune Administrator** (or the scoped

roles above) to the operator. See global requirement G1.

R2. The standard clinical-workstation device group

Description: A Microsoft Entra **device** group `EPIC – Clinical Workstations` containing all in-scope clinical endpoints, and **per-unit device groups** for printer/default-printer mapping — named `EPIC – Clinical Workstations – <UNIT>` (e.g. `EPIC – Clinical Workstations – ED`, `EPIC – Clinical Workstations – 4West`). **Why required:** The baseline profiles assign to `EPIC – Clinical Workstations`; the **printer-deployment policy and per-unit default printer** assign to the per-unit groups, because the default printer and the printer set differ by unit. **How to verify:** Entra → **Groups** → **All groups**; the population group and one group per in-scope unit exist with intended membership. **How to implement if missing:** Create them in Entra (or run `EPICWS-Set-PrinterDeployment.ps1`, which can ensure the per-unit groups). Use dynamic membership rules where a reliable device attribute (name prefix per unit) exists.

R3. Printer delivery method decided — Universal Print vs scripted install

Description: A decision, per printer class, of **how** printers reach the workstation: **Universal Print** (cloud-native, no on-prem print server) or **scripted printer installation** (direct-IP / existing print-server queues via a platform script). **Why required:** The two paths have different prerequisites and different Intune objects; the implementation steps and the `EPICWS-Set-PrinterDeployment.ps1` mode depend on which you choose. **How to verify:** Confirm which is true today: are printers shared from an on-prem print server / direct IP (scripted install), or registered in Universal Print? Check **Intune** → **Tenant administration** → **Connectors and tokens** → **Universal Print** and your print-server console. **How to implement if missing:** Choose per printer class. For Universal Print, satisfy R4. For scripted install, confirm the print server / printer IPs and drivers (R5, R6).

R4. Universal Print licensing and registration (only if using Universal Print)

Description: Universal Print entitlement (included in M365 E3/E5 and equivalents), the Universal Print **printer share(s)** registered, and — for non-native printers — the **Universal Print connector** installed against the print queues. **Why required:** Intune's **Printers** → **Universal Print** deployment only targets printers that are registered and shared in Universal Print; without a license/connector there is nothing to deploy. **How to verify:** Universal Print portal (`https://portal.azure.com` → **Universal Print**) lists the printer(s) as **Registered/Shared**; Intune → **Devices** → **… Universal Print** can select them. **How to implement if missing:** Assign Universal Print–bearing licenses; register native printers or deploy the **Universal Print connector** for legacy/IP printers; share each printer. Skip this requirement entirely if you use scripted install for that printer class.

R5. Print queues / printer endpoints and drivers (only if using scripted install)

Description: The reachable **print-server share paths** (`\\<PRINTSERVER>\<SHARE>`) or **direct-IP** addresses and ports of the wristband/label, script, and document printers, plus the **signed printer drivers** (or the printer's built-in driver) packaged for unattended install. **Why required:** Scripted printer installation adds these named queues and installs the matching driver; without the exact paths/IPs and drivers the script cannot create the printer. **How to verify:** From the test device, reach the share / ping the printer IP

and confirm the driver installs unattended once by hand; run `EPICWS-Discovery.ps1` to inventory currently installed printers and drivers. **How to implement if missing:** Obtain the queue paths / IPs from your print team and the signed drivers from the vendor; stage the driver in the Win32/script package. Record them for the Implementation Guide.

R6. Clinical peripheral drivers/apps — scanners, signature pads, badge/proximity readers

Description: The vendor **drivers/applications** for the in-scope clinical peripherals — document **scanners**, **signature pads** (registration/consent), and **badge / proximity readers** — packaged for unattended deployment via Intune (Win32 app or platform script), with their **detection** signal. **Why required:** Signature pads and many badge readers need a vendor driver/runtime before Epic (or the OS / Imprivata) recognizes them; standardizing means deploying these the same way on every device. **How to verify:** On the test device confirm each peripheral is recognized (Device Manager shows no unknown device; the signature pad / badge reader responds); run `EPICWS-Validation.ps1` which checks for the expected peripheral driver/service. **How to implement if missing:** Obtain signed vendor drivers/installers, wrap them as Intune Win32 apps with silent switches and a detection rule, and assign to `EPIC – Clinical Workstations`. For badge-tap SSO behavior beyond OS recognition, engage the separate Imprivata solution.

R7. Epic print mapping values — from Epic Technical Services / UserWeb

Description: The **Epic Print Service (EPS)** host, the **Epic print queue names**, and the **workflow-to-printer mapping** (which Epic queue a wristband, a script, and a document route to per department), plus the intended **default printer per unit**. **Why required:** These define which physical/Universal Print printer each Epic workflow targets and which printer is the per-unit default; they are **Epic-specific and customer-only** and must not be guessed. **How to verify:** Confirm with your **Epic Technical Services / client systems team** (sourced from **Epic UserWeb**) the EPS host, the exact queue names, and the per-department/per-unit printer mapping. Print a test wristband, script, and document on the test device and confirm each lands on the intended physical printer. **How to implement if missing:** Request these from your Epic team / Epic UserWeb and record them as the placeholders `<EPS_SERVER>`, `<EPIC_QUEUE_WRISTBAND>`, `<EPIC_QUEUE_SCRIPT>`, `<EPIC_QUEUE_DOC>`, and `<UNIT_DEFAULT_PRINTER>` used throughout the Implementation Guide.

R8. Windows security baseline (apply alongside this baseline)

Description: Your organization's approved **Windows security baseline** (Intune security baseline or settings-catalog equivalent), applied to clinical workstations. **Why required:** This solution's baseline tunes power/display/timeout and the clinical experience; it **references** but does not redefine the security baseline. Both must be present and must not conflict. **How to verify:** Intune → **Endpoint security** → **Security baselines** (or your settings-catalog security profile) is assigned to `EPIC – Clinical Workstations` and reports compliant; confirm no setting collides with this solution's power/timeout settings. **How to implement if missing:** Deploy the approved security baseline first, then layer this solution's standardization profile, resolving any setting conflicts in favor of the security baseline. See global requirement G2/G4.

R9. Target operating system — Windows 10 / Windows 11

Description: Clinical endpoints run a supported, patched **Windows 10 (21H2+)** or **Windows 11**, x64. **Why required:** The settings catalog, Universal Print, and modern driver delivery require a current, supported Windows build; profile **applicability/filter rules** will target these OS versions. **How to verify:** `winver` on a device, or Intune → device → **Hardware**. **How to implement if missing:** Patch via Update rings or upgrade devices. See global requirement G2.

R10. A test clinical device (not in patient care), in a representative unit

Description: A representative, enrolled, non-production clinical device (ideally one fixed workstation and one WOW) in a representative **unit**, added to the pilot group. **Why required:** The baseline, printer order, per-unit default printer, and peripheral recognition must be proven before any device in care receives them. **How to verify:** The device is enrolled, compliant, in the pilot group, and in (or representing) the target unit's per-unit group. **How to implement if missing:** Set aside the device and add it to the pilot and per-unit groups. See global requirement G7.

R11. Approved clinical change window and rollback plan

Description: A change-control approval, an agreed **clinical maintenance window** for the expand phase, and a documented **rollback** (revert to the previously validated profile + printer mapping). **Why required:** Standardizing and re-mapping printers touches patient-care devices; it must be change-managed, windowed, and reversible. **How to verify:** A CAB-approved change ticket exists with window and back-out plan, naming the previously validated profile and per-unit default printer to revert to. **How to implement if missing:** Route through your CAB; keep the prior validated profile export and printer mapping for rollback. See global requirements G4, G5, G8.

R12. Imprivata present (only if badge readers feed SSO / badge-tap)

Description: If your clinical endpoints use **Imprivata** for single sign-on / badge-tap, confirm the Imprivata agent is installed and healthy on the test device, so that the **badge/proximity reader** this solution deploys is consumed correctly. **Why required:** This solution makes the OS recognize the badge reader; Imprivata is what turns a tap into a logon. You must confirm the reader is recognized and tap-to-launch still works after the peripheral deployment. (Imprivata configuration itself is a separate solution.) **How to verify:** On the test device, confirm the badge reader enumerates, the Imprivata agent service is running, and a badge tap behaves as expected (Testing §, Test 6). **How to implement if missing:** Engage the separate Imprivata solution / your Imprivata administrators. If you do **not** use Imprivata, this requirement is N/A for SSO — the reader is still deployed and validated as a recognized device.

Requirements checklist (use before standardizing)

- **R1** Intune configuration + app rights (Intune Administrator / scoped roles)
- **R2** `EPIC – Clinical Workstations` + per-unit groups `EPIC – Clinical Workstations – <UNIT>`
- **R3** Printer delivery method decided per class (Universal Print vs scripted install)
- **R4** Universal Print licensed + printers registered/shared (if using Universal Print)
- **R5** Print queue paths / IPs + signed drivers (if using scripted install) — `EPICWS-Discovery.ps1`

• **R6** Peripheral drivers/apps packaged (scanner, signature pad, badge reader) — `EPICWS-Validation.ps1`

• **R7** Epic EPS host + queue names + workflow/printer mapping + per-unit default from Epic TS/UserWeb

• **R8** Windows security baseline applied alongside this baseline

• **R9** Supported, patched Windows 10/11 (x64)

• **R10** Test clinical device (fixed + WOW) in a representative unit + pilot group

• **R11** CAB-approved change window + rollback to prior validated profile/mapping

• **R12** Imprivata agent healthy on test device (only if badge readers feed SSO)

3. Implementation Guide — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune

This guide is ordered. Follow the phases in sequence: **Assess** → **Standardize** → **Deploy Peripherals/Printing** → **Pilot** → **Validate** → **Expand.** Every step gives the **full Intune admin center navigation path** and names every object to a fixed convention so the fleet stays self-documenting. Do not skip the requirements check (**02-requirements.md** + **global-requirements-healthcare-solutions.md**) before starting.

> ***Named objects (reused verbatim throughout):*** - ***Device group (population):*** `EPIC – Clinical Workstations` - ***Per-unit device groups:*** `EPIC – Clinical Workstations – <UNIT>` - ***Standardization profile:*** `EPIC – Clinical Workstations – Baseline` *(settings catalog)* - ***Power/display profile:*** `EPIC – Clinical Workstations – Power & Display` - ***Printer-deployment policy:*** `EPIC – Clinical Workstations – Printers – <UNIT>` - ***Default-printer setting:*** `EPIC – Clinical Workstations – Default Printer – <UNIT>` - ***Peripheral apps:*** `EPIC – Peripheral – Signature Pad`, `EPIC – Peripheral – Document Scanner`, `EPIC – Peripheral – Badge Reader` ***Epic-specific values*** *(*`<EPS_SERVER>`, `<EPIC_QUEUE_WRISTBAND>`, `<EPIC_QUEUE_SCRIPT>`, `<EPIC_QUEUE_DOC>`, `<UNIT_DEFAULT_PRINTER>`*) and* ***print endpoints*** *(`\\<PRINTSERVER>\<SHARE>` / direct-IP) come from your* ***Epic team / print team*** *— substitute them; never invent them.*

Phase A — Assess

Step 1 — Confirm prerequisites and capture the baseline state

Description: Verify the requirements and record the current state of a representative device. **Why:** You cannot prove standardization worked without a documented "before." **Verify / Implement:** Walk **02-requirements.md** R1–R12. On the test device run `EPICWS-Discovery.ps1` to inventory installed printers, default printer, peripheral drivers/services, and current power/timeout settings. Save the output as the pre-change baseline and attach it to the change ticket (G4).

Step 2 — Confirm the device groups exist

Description: Ensure the population group and the per-unit groups exist. **Why:** Baseline profiles target the population group; printers and the default printer target the per-unit groups. **Implement:** Intune admin center (`https://intune.microsoft.com`) → **Groups** → **All groups** → confirm `EPIC – Clinical Workstations` and `EPIC – Clinical Workstations – <UNIT>` for each in-scope unit. If missing, **+ New group** → **Security**, membership type **Dynamic Device** (rule on the unit's device-name prefix) or **Assigned**, enter the name and description, **Create**. (R2)

Step 3 — Decide and record printer delivery method per printer class

Description: For wristband/label, script, and document printers, decide **Universal Print** or **scripted install** (R3). **Why:** The objects you create in Phase C differ by method. **Implement:** Record the decision per class. If Universal Print, confirm registration: **portal.azure.com** → **Universal Print** → **Printers** shows each printer **Shared** (R4). If scripted, confirm the share path / IP and the signed driver are ready (R5).

Phase B — Standardize (the device baseline)

Step 4 — Create the standardization profile (settings catalog)

Description: Build the baseline device-experience profile. **Why:** This is the single source of the consistent clinical-workstation experience. **Implement:**

1. Intune admin center → **Devices** → **Configuration** → **+ Create** → **New Policy**.
2. **Platform:** Windows 10 and later. **Profile type: Settings catalog**. **Create**.
3. **Name:** `EPIC – Clinical Workstations – Baseline`. **Description:** "Standard clinical workstation

experience for Epic — power/display/lock behavior. Managed by the standard deployment model." **Next.**

1. **Configuration settings** → **+ Add settings.** Add the agreed clinical settings (examples — confirm

against your security baseline, R8): screen-lock / inactivity timeout aligned to clinical policy, notification/UX suppression, and any required local-experience settings. Do **not** duplicate settings already owned by the security baseline.

1. **Next** through Scope tags → **Assignments**: do not assign yet (Phase D pilots first). **Next →

Create.**

Step 5 — Create the power & display profile

Description: Tune power/sleep/display so always-on shared clinical devices do not sleep mid-shift. **Why:** A workstation that sleeps or turns off the display in a patient room interrupts care and generates tickets; this must be identical across the fleet. **Implement:**

1. **Devices** → **Configuration** → **+ Create** → **New Policy** → Windows 10 and later → **Settings catalog**

→ **Create**.

1. **Name:** `EPIC – Clinical Workstations – Power & Display`. **Description** as above. **Next.**
2. **+ Add settings** → search **Power** → set, for plugged-in clinical workstations: turn off

display after the clinically agreed interval (not 0/never unless policy requires), **never sleep**, disable hibernate, and the agreed lid/idle behavior for WOWs. **Next.**

1. Scope tags → **Assignments**: leave unassigned for now → **Next** → **Create.**

Step 6 — Confirm the security baseline is applied (do not redefine it)

Description: Verify the Windows security baseline is present alongside this baseline (R8). **Why:** This solution layers experience settings on top of, never instead of, security. **Implement: Endpoint security** → **Security baselines** → open your assigned baseline → confirm it targets `EPIC – Clinical Workstations` and reports compliant. Resolve any conflict with Steps 4–5 in favor of the security baseline. If missing, deploy it first.

Phase C — Deploy Peripherals / Printing

Step 7 — Deploy clinical peripheral drivers/apps

Description: Make every workstation recognize the scanner, signature pad, and badge reader (R6). **Why:** Standardization means these peripherals are installed and validated the same way everywhere. **Implement (per peripheral):**

1. Intune admin center → **Apps** → **Windows** → **+ Add** → app type **Windows app (Win32)**.
2. Upload the vendor's `.intunewin` (wrap the signed driver/installer with the Microsoft Win32 Content Prep Tool first). **Next.**
1. **Name** the app exactly: `EPIC – Peripheral – Signature Pad` (repeat for `EPIC – Peripheral – Document Scanner`, `EPIC – Peripheral – Badge Reader`). **Next.**
1. **Program:** silent **Install command** and **Uninstall command** from the vendor. **Next.**
2. **Requirements:** Windows 10/11, x64 (R9). **Next.**
3. **Detection rules:** the vendor's installed file/version or registry key (the value `EPICWS-Validation.ps1` checks). **Next** through Dependencies/Supersedence.
1. **Assignments** → **Required** → **+ Add group** → `EPIC – Clinical Workstations`. **Next** → **Create.**

Step 8 — Deploy printers via Universal Print (if chosen, R3/R4)

Description: Push the registered Universal Print printers to each unit's devices. **Why:** Universal Print gives a cloud-native printer deployment with no print server. **Implement (per unit):**

1. Intune admin center → **Devices** → **Windows** → **(Manage devices)** → **Configuration** → **+ Create** →

Windows 10 and later → **Templates** → **Universal Print printer** → **Create**. *(If your tenant surfaces it under **Apps** → **Universal Print** or **Printers**, use that blade — the object is the same.)*

1. **Name:** `EPIC – Clinical Workstations – Printers – <UNIT>`. **Description:** "Unit printer set —

wristband/script/document — Universal Print. Standard deployment model." **Next.**

1. **+ Add printer** → select the unit's **wristband/label**, **script**, and **document** printers

(the Universal Print shares mapped to <EPIC_QUEUE_WRISTBAND> / <EPIC_QUEUE_SCRIPT> / <EPIC_QUEUE_DOC> per R7). **Next.**

1. Scope tags → **Assignments** → **+ Add group** → EPIC – Clinical Workstations – <UNIT>. **Next →

Create.** Repeat per unit.

Step 9 — Deploy printers via scripted install (if chosen, R3/R5)

Description: Install the unit's printers from print-server shares / direct IP with a platform script. **Why:** Scripted install works with existing print servers and direct-IP clinical printers without Universal Print. **Implement (per unit):**

1. Generate the per-unit script with **`EPICWS-Set-PrinterDeployment.ps1 -Unit <UNIT> -Mode Script

-WhatIf (preview), then without -WhatIf to emit the install script naming each queue (\\<PRINTSERVER>\<SHARE> or Add-Printer -PortName IP_<addr>), in wristband → script → document order, and setting <UNIT_DEFAULT_PRINTER> as default.

1. Intune admin center → **Devices** → **Windows** → **Scripts and remediations** → **Platform scripts** → **+

Add → Windows 10 and later**.

1. **Name:** EPIC – Clinical Workstations – Printers – <UNIT>. **Next.**

2. Upload the generated script; **Run this script using the logged-on credentials = No** (run as

system for machine-wide printers); **Enforce script signature check** per your policy; **Run script in 64-bit** = Yes. **Next.**

1. Scope tags → **Assignments** → **+ Add group** → EPIC – Clinical Workstations – <UNIT>. **Next →

Add.** Repeat per unit.

Step 10 — Set the correct default printer per unit

Description: Make the right printer the default for each unit's workstations. **Why:** The default printer is what an Epic document/print dialog targets when the clinician does not choose — it must be the unit's intended default (<UNIT_DEFAULT_PRINTER>, R7), or work routes to the wrong room. **Implement:**

• **Universal Print path:** in the unit's printer-deployment policy (Step 8) mark

<UNIT_DEFAULT_PRINTER> as the **default**.

• **Scripted path:** the generated script (Step 9) already issues the default-printer command; confirm

it names <UNIT_DEFAULT_PRINTER>.

• Either way create/confirm the tracking object name `EPIC – Clinical Workstations – Default Printer –

<UNIT>` so the mapping is documented per unit.

Phase D — Pilot

Step 11 — Assign everything to the pilot

Description: Assign the baseline, power/display, peripherals, and one unit's printers to the pilot device only. **Why:** Prove the full standard on a non-production device (G7, R10) before any device in care receives it. **Implement:**

1. For `EPIC – Clinical Workstations – Baseline` and `EPIC – Clinical Workstations – Power & Display`:

Devices → **Configuration** → open each → **Properties** → **Assignments** → **Edit** → add a **pilot** group (or add only the test device to `EPIC – Clinical Workstations` initially). **Review + save.**

1. Confirm the peripheral apps (Step 7) and the pilot unit's printer policy (Step 8/9) reach the test

device.

1. On the test device run `gpupdate`-equivalent sync: **Settings → Accounts → Access work or school →

Info → Sync**, or in Intune open the device** → Sync**.

Step 12 — Wait for apply and observe

Description: Let policies, apps, and printers land; watch for failures. **Why:** Catch errors on one device, not the floor. **Implement:** Intune → **Devices** → **All devices** → test device → **Device configuration** (profiles **Succeeded**), **Managed Apps** (peripheral apps **Installed**), and the printer policy status. Investigate anything not **Succeeded/Installed** before validating.

Phase E — Validate

Step 13 — Run the validation gate on the pilot device

Description: Confirm the standard actually landed. **Why:** Validation is the gate; do not expand on a partial result. **Implement:** On the pilot device run `EPICWS-Validation.ps1 -Unit <UNIT> -ExpectedDefault "<UNIT_DEFAULT_PRINTER>"`. It checks: baseline/power settings applied, the unit's wristband/script/document printers present in the right order, the **default printer** equals `<UNIT_DEFAULT_PRINTER>`, and the scanner / signature pad / badge reader recognized. Then perform the clinical print tests in **05-testing.md** (a real wristband, script, and document from Epic land on the intended physical printers, per <EPS_SERVER> mapping). The overall verdict must be **PROVISIONED** with all clinical tests PASS before Phase F.

Step 14 — Capture the validated state as the rollback reference

Description: Record the proven configuration for back-out (G8, R11). **Why:** A clinical change must be reversible. **Implement:** Export the two settings-catalog profiles (**…** → **Export JSON** where available, or record settings), and run `EPICWS-Report.ps1` to CSV for the pilot. File both with the change ticket as the "known-good" reference.

Phase F — Expand

Step 15 — Expand to the rest of the unit, then unit by unit

Description: Broaden assignment from the pilot to full units, in the change window. **Why:** Controlled expansion contains risk; each unit needs its own default printer and printer set. **Implement:**

1. In your approved **maintenance window** (G5, R11), change the baseline / power profile assignments to target the full `EPIC – Clinical Workstations` group.

1. For each additional unit, repeat Steps 8–10 to create `EPIC – Clinical Workstations – Printers – <UNIT>` and the unit default, assigned to `EPIC – Clinical Workstations – <UNIT>`.

1. After each unit, re-run `EPICWS-Validation.ps1 -Unit <UNIT>` on a sample device and a clinical print test before moving to the next unit.

Step 16 — Operationalize

Description: Move from project to steady state. **Why:** Standardization only holds if drift is monitored. **Implement:** Schedule `EPICWS-Report.ps1` (CSV) periodically to detect drift (wrong default printer, missing peripheral, profile not applied); triage with `EPICWS-Remediation.ps1` (read-only / `-WhatIf` first, `-Apply` in a window). See **07-best-practices.md**.

> ***Reminder:*** *the Epic Print Service host, queue names, and workflow/printer mapping (`<EPS_SERVER>`, `<EPIC_QUEUE_*>`, `<UNIT_DEFAULT_PRINTER>`) come from your Epic team / Epic UserWeb. The Intune standardization, printer-deployment (Universal Print vs scripted), peripheral deployment, and validation mechanics above are exact; substitute the Epic/print values consistently.*

5. Testing & Validation — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune

This is the **acceptance gate**. Run every test on the **pilot device** (Implementation Phase E) before expanding, and re-run the unit-specific tests on a sample device after each unit in Phase F. Each test has an explicit **PASS / FAIL** condition. The rollout may expand only when **all tests PASS** (or are justified N/A).

> *The PowerShell tool `EPICWS-Validation.ps1` automates Tests 1–5; the **clinical print tests** (Tests 7–9) and the **Imprivata tap test** (Test 6) require a person at the device because they confirm an Epic workflow lands on a physical printer. Tools report device/printer state only — never patient data (G4).*

Test 1 — Standardization baseline applied

Goal: The `EPIC – Clinical Workstations – Baseline` settings-catalog profile applied. **Steps:** Intune → **Devices** → **All devices** → device → **Device configuration**; and run `EPICWS-Validation.ps1`. **PASS:** Profile status **Succeeded**; the validation script reports the baseline settings present. **FAIL:** Profile **Error/Conflict**, or settings not present → resolve conflict (often the security baseline, R8) and re-sync.

Test 2 — Power & display behavior correct

Goal: The workstation does not sleep / turn off display contrary to clinical policy. **Steps:** Confirm `EPIC – Clinical Workstations – Power & Display` is **Succeeded**; on the device check `powercfg /q` (or the script's power read) for never-sleep and the agreed display timeout. **PASS:** Sleep disabled, hibernate disabled, display timeout = the clinically agreed value. **FAIL:** Device sleeps or display times out off-policy → correct the profile and re-test.

Test 3 — Unit printers present in the correct order

Goal: The wristband/label, script, and document printers for the unit are installed. **Steps:** Run `EPICWS-Validation.ps1 -Unit <UNIT>`; or `Get-Printer` on the device. **PASS:** All three printer classes present, named to the unit's mapping, in **wristband** → **script** → **document** order. **FAIL:** A printer missing or mis-ordered → re-check the Step 8/9 deployment for that unit and re-sync.

Test 4 — Default printer equals the unit default

Goal: The default printer is `<UNIT_DEFAULT_PRINTER>` for this unit. **Steps:** Run `EPICWS-Validation.ps1 -Unit <UNIT> -ExpectedDefault "<UNIT_DEFAULT_PRINTER>"`; or `Get-CimInstance Win32_Printer | ? Default`. **PASS:** Default printer = `<UNIT_DEFAULT_PRINTER>`. **FAIL:** Wrong default → re-apply Step 10 for the unit and re-test.

Test 5 — Clinical peripherals recognized

Goal: The scanner, signature pad, and badge/proximity reader are installed and recognized. **Steps:** Run `EPICWS-Validation.ps1` (peripheral driver/service checks); confirm Device Manager shows no unknown device for these peripherals. **PASS:** Each in-scope peripheral's driver/service present; no unknown device. **FAIL:** A peripheral unrecognized → confirm its Win32 app shows **Installed** (Step 7) and the device is connected; re-deploy/re-test.

Test 6 — Badge reader feeds SSO (only if using Imprivata, R12)

Goal: The deployed badge/proximity reader is consumed by Imprivata for tap behavior. **Steps:** With the Imprivata agent running, tap an enrolled badge at the device. **PASS:** The tap behaves as configured (logon / fast-user-switch) and tap-to-launch into Epic works. **FAIL:** Reader recognized (Test 5) but tap does nothing → engage the separate Imprivata solution. If not using Imprivata, mark **N/A** (Test 5 still governs reader recognition).

Test 7 — Epic wristband prints to the wristband printer

Goal: A real Epic wristband print job lands on the wristband/label printer. **Steps:** From Epic on the device, print a test wristband to the workflow's wristband queue (`<EPIC_QUEUE_WRISTBAND>` via `<EPS_SERVER>`). **PASS:** The wristband prints on the unit's **wristband/label** printer, correctly formatted. **FAIL:** Wrong printer or no output → re-check the Epic queue→printer mapping (R7) with your Epic team and the printer deployment.

Test 8 — Epic script/prescription prints to the script printer

Goal: A script/prescription job lands on the script printer. **Steps:** From Epic, print a test script to `<EPIC_QUEUE_SCRIPT>`. **PASS:** The script prints on the unit's **prescription/script** printer. **FAIL:** Wrong

printer → re-verify mapping (R7) and deployment; re-test.

Test 9 — Epic document prints to the document printer / default

Goal: A general document job lands on the document printer (the unit default). **Steps:** From Epic, print a test document to <EPIC_QUEUE_DOC> (and a no-queue-specified print to confirm it uses the default). **PASS:** The document prints on the unit's **document** printer; default-routed jobs use <UNIT_DEFAULT_PRINTER>. **FAIL:** Wrong printer/default → re-verify Steps 8–10 and mapping; re-test.

Test 10 — Repeatability across form factors

Goal: A fixed workstation **and** a WOW in the same unit produce identical results. **Steps:** Repeat Tests 1–9 on one of each form factor. **PASS:** Both devices present the same baseline, printers, default, and recognized peripherals. **FAIL:** Divergence → reconcile group membership/assignment so both are in the same unit/population groups; re-test.

Acceptance gate (must hold before Phase F expansion)

- **T1** Baseline profile **Succeeded** / settings present
- **T2** Power/display per clinical policy (never-sleep, agreed display timeout)
- **T3** Unit printers present, correct order
- **T4** Default printer = <UNIT_DEFAULT_PRINTER>
- **T5** Scanner + signature pad + badge reader recognized
- **T6** Badge tap feeds SSO (or N/A)
- **T7** Wristband → wristband printer
- **T8** Script → script printer
- **T9** Document → document printer; default-routed → unit default
- **T10** Fixed workstation and WOW identical

`EPICWS-Validation.ps1` must report **PROVISIONED**, and Tests 6–10 must PASS (or be justified N/A), before any device in patient care is added in Phase F.

6. Troubleshooting — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune

Symptom-driven fixes for the standardization, printer-deployment, default-printer, and peripheral layers. Each entry: **Symptom** → **Likely cause** → **Fix.** Start with read-only tools (`EPICWS-Discovery.ps1`, `EPICWS-Validation.ps1`, `EPICWS-Report.ps1`); only `EPICWS-Remediation.ps1` and `EPICWS-Set-PrinterDeployment.ps1` change anything, and only with `-Apply` in a maintenance window.

Standardization baseline

Symptom: The baseline profile shows Error or Conflict in Intune. Likely cause: A setting in `EPIC – Clinical Workstations – Baseline` or `… – Power & Display` collides with the Windows security baseline (R8) or another profile. **Fix:** Intune → **Devices** → **Configuration** → **(profile)** → **Device/Per-setting status** to find the conflicting setting. Remove the duplicate from this solution's profile in favor of the security baseline; re-sync (`EPICWS-Remediation.ps1`) and re-check (Test 1).

Symptom: Profile reports Succeeded but settings aren't actually in effect on the device. Likely cause: Pending reboot, or the device hasn't synced since assignment. **Fix:** Run `EPICWS-Remediation.ps1 -Apply` to trigger a sync; reboot in a window if a setting requires it; re-validate.

Power / display (device sleeps in a patient room)

Symptom: A clinical workstation sleeps or the display turns off too soon. Likely cause: `EPIC – Clinical Workstations – Power & Display` not applied, or another power profile/GPO is winning. **Fix:** Confirm the power profile is **Succeeded** (Test 2). On the device, `EPICWS-Discovery.ps1` shows the active power indices. If a legacy GPO sets power, remove it for clinical OUs. Re-assign and re-test.

Printers — presence and order

Symptom: One of the unit's printers (wristband/script/document) is missing. Likely cause: The device is not in the unit group `EPIC – Clinical Workstations – <UNIT>`, the printer-deployment policy didn't apply, or (scripted) the share/IP was unreachable. **Fix:** Confirm group membership (`EPICWS-Report.ps1` flags devices with no unit group). Confirm the policy `EPIC – Clinical Workstations – Printers – <UNIT>` is assigned and **Succeeded**. For scripted install, verify the device can reach `\\<PRINTSERVER>\<SHARE>` / the printer IP, and that the signed driver installs unattended (R5). Restart the spooler with `EPICWS-Remediation.ps1 -Apply -RestartSpooler`, re-sync, re-test (Test 3).

Symptom: Printers appear in the wrong order. Likely cause: Two deployment sources (Universal Print policy + a script) both add printers. **Fix:** Pick one delivery method per printer class (R3). Remove the duplicate source; redeploy via `EPICWS-Set-PrinterDeployment.ps1` so the order is wristband → script → document.

Default printer wrong

Symptom: The default printer is not `<UNIT_DEFAULT_PRINTER>` **(Epic "just print" goes to the wrong room). Likely cause:** No default was set, the user/Windows "let Windows manage my default printer" is on, or two sources fight over the default. **Fix:** On the device, ensure **Settings** → **Bluetooth & devices** → **Printers & scanners** → **Let Windows manage my default printer = Off** (set this in the baseline profile to standardize it). Re-assert the default with `EPICWS-Remediation.ps1 -Apply -SetDefaultPrinter "<UNIT_DEFAULT_PRINTER>"`. Ensure only one source (Step 10) owns the default per unit. Re-test (Test 4).

Universal Print vs scripted-install issues

Symptom: Universal Print printer won't deploy / isn't selectable in the policy. Likely cause: Printer not registered/shared in Universal Print, missing license, or the connector isn't running for a legacy printer (R4). **Fix:** portal.azure.com → **Universal Print** → **Printers**: confirm **Shared**; confirm Universal Print licensing; for legacy printers confirm the **Universal Print connector** is healthy. Then re-select the printer in `EPIC – Clinical Workstations – Printers – <UNIT>`.

Symptom: Scripted printer-install platform script fails or installs no printer. Likely cause: Run context (logged-on vs system), 32/64-bit, unreachable share/IP, or missing driver. **Fix:** Intune → **Scripts and remediations** → **Platform scripts** → **(script)** → **Device status** for the error. Confirm **run as system**, **64-bit = Yes**, share/IP reachable, and the driver name in the generated script matches a signed driver present on the device. Regenerate with `EPICWS-Set-PrinterDeployment.ps1` and the correct values.

Clinical peripherals

Symptom: Signature pad / scanner / badge reader not recognized. Likely cause: The peripheral's Intune Win32 app didn't install, the detection rule is wrong, or the device isn't physically connected. **Fix:** Intune → **Apps** → **(peripheral app)** → **Device install status**: confirm **Installed**. If "Installed" but unrecognized, the detection rule may be false-positive — verify the real driver file/ registry value and fix the rule (Step 7). `EPICWS-Discovery.ps1` lists driver-problem devices and recognized peripherals; reconnect the hardware and re-test (Test 5).

Symptom: Badge reader is recognized but a tap does nothing. Likely cause: The reader (OS recognition) is this solution's job; the **tap behavior** is Imprivata's (R12) — the boundary between Test 5 and Test 6. **Fix:** Confirm the Imprivata agent service is running; engage the separate Imprivata solution to fix tap/SSO. If you don't use Imprivata, the reader being recognized (Test 5) is the expected end state.

Epic mapping mismatches

Symptom: Wristband/script/document prints to the wrong physical printer from Epic. Likely cause: The Epic queue→printer mapping (EPS) doesn't match the printer this solution deployed — an Epic-side configuration value, not an Intune one. **Fix:** This is owned by your **Epic Technical Services team / Epic UserWeb** (R7). Confirm `<EPIC_QUEUE_WRISTBAND>` / `<EPIC_QUEUE_SCRIPT>` / `<EPIC_QUEUE_DOC>` on `<EPS_SERVER>` point at the printers you deployed for the unit. Do not change Epic-internal values yourself — coordinate with the Epic team, then re-run the clinical print tests (Tests 7–9).

Drift and expansion

Symptom: A re-imaged device lost its standard (no printers / wrong default). Likely cause: It fell out of `EPIC – Clinical Workstations` and/or its unit group, so no profiles or printers target it. **Fix:** Use dynamic device groups keyed on the unit name prefix so re-imaged devices re-join automatically; otherwise re-add the device to the population and unit groups. `EPICWS-Report.ps1` surfaces devices missing a unit group. Re-sync and re-validate.

Symptom: A whole unit is inconsistent after expansion. Likely cause: The unit's printer policy / default object was never created for that unit. **Fix:** Repeat Implementation Steps 8–10 to create `EPIC – Clinical Workstations – Printers – <UNIT>` and the unit default for the missed unit, assigned to its unit group; validate a sample device (Test 10) before declaring the unit done.

7. Best Practices — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft Intune

Operating practices that keep clinical workstations standardized, printing deterministic, and changes safe in a live clinical environment. They assume the named objects and phase model from **03-implementation.md**.

Naming and self-documentation

- **Reuse the naming convention verbatim**: `EPIC – Clinical Workstations` (population), `EPIC – Clinical Workstations – <UNIT>` (per-unit), `EPIC – Clinical Workstations – Baseline`, … `– Power & Display`, … `– Printers – <UNIT>`, … `– Default Printer – <UNIT>`, and `EPIC – Peripheral – <Device>`. The pattern `<Purpose> – <Scope> – <Stage>` makes the fleet auditable at a glance.
- **One profile, one purpose.** Keep baseline experience separate from power/display and separate from the security baseline. Mixed-purpose profiles are hard to roll back cleanly.

Standardize through groups, not per-device clicks

- Bind everything to **device groups**; never hand-configure a printer or setting on an individual clinical workstation. The whole value of the standard is that it is reproducible.
- Prefer **dynamic device groups** keyed on a reliable unit naming prefix so a re-imaged device lands back in the right unit (and gets the right default printer) automatically.

Printer delivery — choose deliberately, document the choice

- **Universal Print** when you want cloud-native delivery with no print server; budget for the licensing and either native Universal Print printers or the connector. **Scripted install** when you must work with existing print servers / direct-IP clinical printers. Record which method each printer class uses (R3) so troubleshooting starts from the right place.
- Deploy printers in a **deterministic order** — wristband → script → document — so the printer list a clinician sees is identical on every device in the unit.

The default printer is a patient-safety control

- A wrong default printer routes a clinician's "just print" to the wrong room. Treat `<UNIT_DEFAULT_PRINTER>` as a controlled value per unit, validate it (Test 4) on every device, and monitor for drift with `EPICWS-Report.ps1`.
- Never let a Universal Print policy and a scripted script **both** set a default on the same device — pick one owner of the default per unit.

Peripherals: package once, deploy everywhere

• Wrap each peripheral (signature pad, scanner, badge reader) as a **named Win32 app with a real

detection rule** so Intune reports true install state. A peripheral that "works on the bench" but has no detection rule will silently fail to standardize.

• For **badge/proximity readers**, deploying the reader driver (OS recognition) and configuring

Imprivata (tap behavior) are two separate jobs — keep them in their own solutions and validate the boundary (Test 5 vs Test 6).

Respect the Epic / print-team boundary

• Never invent the EPS host, Epic queue names, or workflow→printer mapping. Source them from **Epic

Technical Services / UserWeb** (R7) and substitute the placeholders consistently. Your job is the Intune + printer + peripheral mechanics that carry those values onto every device correctly.

Clinical-environment safety

• **Pilot before expand, always.** Prove the full standard (baseline + printers + default +

peripherals) on a non-production device before any device in care receives it.

• **Change windows and rollback.** Make assignment changes in approved maintenance windows (G5); keep

the exported profile and prior per-unit default printer as the named rollback reference (G8) before each change.

• **Read-only first.** Run `EPICWS-Discovery.ps1`, `EPICWS-Validation.ps1`, and `EPICWS-Report.ps1`

freely; run `EPICWS-Remediation.ps1` and `EPICWS-Set-PrinterDeployment.ps1` with `-WhatIf` first and only `-Apply` in a window.

• **No ePHI in tooling output.** The scripts report device/printer/app state only. Keep it that way —

never extend them to read clinical content (G4).

Hold the line against drift

• Schedule `EPICWS-Report.ps1` to CSV on a cadence and review for: profile not applied, missing

printer, wrong default, missing peripheral. Drift is normal on shared clinical devices; catching it early is what keeps the fleet standardized.

• Re-validate after **any** unit change, re-image, or printer-infrastructure change.

8. Deliverables Summary — Epic Clinical Workstation Standardization & Peripheral Configuration with Microsoft

Intune

This solution is delivered as a complete, click-to-implement package: implementation documentation, an operational PowerShell toolkit, and a hero image.

Documentation

Document	Purpose
executive-summary.md	Business-level overview, value, and approach for decision-makers.
01-solution-overview.md	What the solution delivers, devices in scope, scope boundaries, and how to use the package.
02-requirements.md	Standardization/peripheral/printing-specific requirements (R1–R12), each with Description / Why / Verify / Implement, plus a checklist.
global-requirements-healthcare-solutions.md	Category-wide prerequisites: tenant + roles, Windows licensing, HIPAA/HITRUST change control, clinical maintenance windows, network (incl. Universal Print + print servers + EPS), test device, rollback.
03-implementation.md	Ordered, full-navigation steps: standardize the baseline + power/display, deploy peripherals, deploy printers (Universal Print or scripted) with a per-unit default, pilot, validate, then expand.
05-testing.md	Numbered tests with PASS/FAIL gates — baseline, power, printer order, default printer, peripherals, and live Epic wristband/script/document print tests — plus the acceptance gate.
07-best-practices.md	Operating practices for naming, group-driven standardization, Universal-Print-vs-scripted choice, the default printer as a safety control, peripheral packaging, and clinical-environment safety.
06-troubleshooting.md	Fixes for profile conflicts, devices that sleep, missing/mis-ordered printers, wrong default printer, Universal Print vs scripted issues, unrecognized peripherals, badge-reader/SSO boundary, and Epic-mapping mismatches.

PowerShell toolkit (`scripts\`)

Script	What it does
EPICWS-Discovery.ps1	Inventories a device's installed printers, default printer, peripheral drivers/services, and power/timeout settings. Read-only.
EPICWS-Validation.ps1	Verifies the baseline applied, the unit's printers are present in order, the **default printer** equals the unit default, and the scanner/signature pad/badge reader are recognized; emits PASS/FAIL and an overall verdict.
EPICWS-Report.ps1	Produces a fleet standardization/drift report (per-device baseline, default printer, peripheral status) to **CSV**.
EPICWS-Remediation.ps1	Safely re-triggers an MDM sync / re-applies a named profile or app install-state nudge for a device. Read-only / `-WhatIf` by default; `-Apply` to act.
EPICWS-Set-PrinterDeployment.ps1	Generates/maps the per-unit printer deployment (Universal Print selection or a scripted-install script) and the per-unit default printer. Supports `-WhatIf`.

Each script carries a `.SYNOPSIS` plus **PURPOSE / HOW TO RUN / EXPECTED RESULTS** in its header.

Visual asset

Asset	Purpose
EPICWS-hero.png	Hero image for the solution listing.

Named objects you create (reused verbatim across the docs and scripts)

- **Device groups:** `EPIC – Clinical Workstations`, `EPIC – Clinical Workstations – <UNIT>`
- **Configuration profiles:** `EPIC – Clinical Workstations – Baseline`, `EPIC – Clinical Workstations – Power & Display`
- **Printer-deployment policy:** `EPIC – Clinical Workstations – Printers – <UNIT>`
- **Default-printer object:** `EPIC – Clinical Workstations – Default Printer – <UNIT>`
- **Peripheral apps:** `EPIC – Peripheral – Signature Pad`, `EPIC – Peripheral – Document Scanner`, `EPIC – Peripheral – Badge Reader`

> *Epic/print-specific values (`<EPS_SERVER>`, `<EPIC_QUEUE_WRISTBAND>`, `<EPIC_QUEUE_SCRIPT>`, `<EPIC_QUEUE_DOC>`, `<UNIT_DEFAULT_PRINTER>`, and print endpoints `\\<PRINTSERVER>\<SHARE>` / direct-IP) come from your **Epic team / print team** and are substituted consistently throughout.*

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

EPICWS-Discovery.ps1

Purpose: Reports, on the device it runs on, the current state relevant to Epic clinical-workstation standardization: installed printers (and order), the default printer, recognized clinical peripherals (signature pad / document scanner / badge reader, matched by driver/service name), and key power/display/timeout settings. Use during Assess (Implementation Step 1) to capture the pre-change baseline, and any time to inventory a device. Read-only.

How to run: On the device, in a normal PowerShell window: powershell -ExecutionPolicy Bypass -File .\EPICWS-Discovery.ps1 Optionally pass name patterns for your peripheral drivers/services: .\EPICWS-Discovery.ps1 -SignaturePadLike 'Topaz|SigPad' -ScannerLike 'Fujitsu|Scanner' -BadgeReaderLike 'RFIDeas|pcProx|Imprivata'

EPICWS-Remediation.ps1

Purpose: For a device that failed validation (EPICWS-Validation.ps1), safely nudges it back toward the standard: triggers an Intune MDM sync so pending configuration profiles, the unit printer policy, and peripheral apps re-evaluate and re-apply. Optionally restarts the print spooler (a common cause of "printer missing/wrong default" after a deployment) and re-asserts the unit default printer locally. READ-ONLY / -WhatIf by DEFAULT — it previews every action and changes nothing until you pass -Apply, and only ever in an approved maintenance window.

How to run: On the affected device, in an elevated PowerShell window: Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\EPICWS-Remediation.ps1 Apply in a maintenance window: .\EPICWS-Remediation.ps1 -Apply -RestartSpooler -SetDefaultPrinter "<UNIT_DEFAULT_PRINTER>"

EPICWS-Report.ps1

Purpose: Builds a fleet-level report of clinical-workstation standardization across the managed devices, using Microsoft Graph (Intune managed-device data) for inventory, and writes it to CSV for triage and drift detection. Reports each device's unit group membership where available, its compliance/management state, and last-sync — the signals that show whether the standard is holding. For deep per-device printer/default/peripheral checks, run EPICWS-Validation.ps1 on the device. Read-only.

How to run: Requires the Microsoft.Graph SDK and an operator with device-read rights: Install-Module Microsoft.Graph -Scope CurrentUser # one time powershell -ExecutionPolicy Bypass -File .\EPICWS-Report.ps1 ` -GroupName "EPIC - Clinical Workstations" -OutCsv .\EPICWS-fleet-report.csv

EPICWS-Set-PrinterDeployment.ps1

Purpose: Produces the per-unit printer-deployment artifact this solution uses, named to the convention "EPIC - Clinical Workstations - Printers - ": - Mode Script -> emits a signed-ready PowerShell install script that adds the unit's wristband/script/document printers (from print-server shares or direct IP), in order, and sets as default. Upload this as an Intune platform script (Implementation Step 9). - Mode UniversalPrint -> emits a checklist/plan of the Universal Print shares to select for the unit's printer-deployment policy (Implementation Step 8), including which one to mark default. Also (optionally, via Graph) ensures the

per-unit Entra device group "EPIC - Clinical Workstations - " exists. Supports -WhatIf so you can preview every create/emit before it happens. Generates artifacts only; it does not push policy to devices.

How to run: Preview (no changes/files): powershell -ExecutionPolicy Bypass -File .\EPICWS-Set-PrinterDeployment.ps1 -Unit ED -Mode Script -WhatIf Emit a scripted-install script for the ED unit: .\EPICWS-Set-PrinterDeployment.ps1 -Unit ED -Mode Script `-Wristband '\\PRINTSRV\Wristband-ED' -Script '\\PRINTSRV\Script-ED' -Document '\\PRINTSRV\Doc-ED'` -DefaultPrinter '\\PRINTSRV\Doc-ED' -OutDir .\out Emit a Universal Print selection plan + ensure the unit group: .\EPICWS-Set-PrinterDeployment.ps1 -Unit ED -Mode UniversalPrint `-Wristband 'UP-Wristband-ED' -Script 'UP-Script-ED' -Document 'UP-Doc-ED'` -DefaultPrinter 'UP-Doc-ED' -EnsureGroup

EPICWS-Validation.ps1

Purpose: Validates, on a pilot/broad device, that the Epic clinical-workstation standard landed: (1) the device is Intune-managed / Entra-joined, (2) the unit's wristband/script/document printers are present, (3) the DEFAULT printer equals the expected unit default, (4) power/ display is set to never-sleep with the agreed display timeout, and (5) the clinical peripherals (signature pad / scanner / badge reader) are recognized. Emits PASS/FAIL per check and an overall PROVISIONED / NOT PROVISIONED verdict. Run after Implementation Step 12 (pilot) and after each unit/change. Read-only.

How to run: On the device, in a normal PowerShell window. Supply your unit's expected values: powershell -ExecutionPolicy Bypass -File .\EPICWS-Validation.ps1 `-Unit "ED" -ExpectedDefault "<UNIT_DEFAULT_PRINTER>"` -ExpectedPrinters "Wristband-ED","Script-ED","Doc-ED" ` -MaxDisplayTimeoutMinutes 20